

01

Code specifications

Dataflow

Background

Starts with data in NANOAOD format
DY and ttbar are MC
QCD is based on same sign events from data

Fase0
↓
QCD_fase0.cpp
DY_fase0.cpp
Ttjets_fase0.cpp

↓
Produces fase0 files
for each background
used in:

Fase1
↓
nanotry_fase1_QCD.cpp
nanotry_fase1_DY.cpp
nanotry_fase1_jets.cpp

↓
Produces fase1, one
file for each, after
enriched with pileup
protons

→ QCD_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root
→ DY_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root
→ ttJets_2018_UL_skimmed_TauTau_nano_faseltotal-protons_2018.root

final data files with the signal-like selection (opposite sign taus) :
Dados_2018_UL_skimmed_TauTau_nano_faseltotal-
protons_2018.root

Signal

Starts with data in MINIAOD format

↓
Sinal.cpp

→ data/MC corrections for the efficiency,
scale, and resolution of taus, electrons,
muons, and protons

↓
Produces ntuple with format similar as fase1,
includes BSM weighs needed to normalize the
sample for different SM and BSM scenarios

↓
TauTau_sinal_PIC_july_2018.root

Used in plot_m.cpp
Produces normalized plots and the input distributions for
BDT

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (`HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1`)

Code organization Background

Goal: only one file for signal and one file for background
(with parallelization using `rootEnableImplicit`)

File name	reads		Output
DY_fase0	DY 2018 UL.txt		DY 2018 UL skimmed Tau Tau nano
QCD_fase0	QCD 2018 UL.txt dadosluminosidade.txt	Checks if luminosity in event matches with the data from file	QCD 2018 UL skimmed Ta uTau nano
ttJets_fase0	ttJets 2018 UL.txt		ttJets 2018 UL skimmed TauTau nano
nanotry_fase1_DY	DY 2018.txt Name of fase0 output files		
nanotry_fase1_QCD	QCD fase1 PIC.txt Name of fase0 output files		
nanotry_fase1_ttJets	ttJets fase1 PIC.txt Name of fase0 output files		

Code organization
Background-> fase0 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	tt MC	Data-driven QCD
Luminosity block filtering			✓
Extra gen matching	✓	✓	
Generator weight	1.81	0.15	
HLT version	40_Trk1_eta2p1	40_Trk1_eta2p1	35_Trk1_eta2p1

Code organization
Background-> fase1 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	t $\bar{t}$ MC	Data-driven QCD
Scale Factors	Tau ID + energy scale + sys	Tau ID + energy scale	
Systematic Variations	full		
Charge Selection	Opposite Sign (<0)	Opposite Sign (<0)	Same Sign (>0)
Tau and Muon Selection Cuts	pt tau > 100, η < 2.3, pt muon > 35	pt > 100, η < 2.4, pt muon > 35	pt > 100, η < 2.4, pt muon > 35
Gen-Match Handling	yes		

02

Variables and distributions

μ T channel

Cuts applied

5.2 $\mu\tau_h$ decay mode

This final state is characterised by one tau decaying hadronically and the other one decaying leptonically: $\tau \rightarrow \nu_\tau \mu \bar{\nu}_\mu$. The list of the cuts applied is:

- one isolated muon and one isolated τ_h in the final state (identified using the ID algorithms mentioned above);
- τ_h and muon must have charge of opposite sign (to satisfy the charge conservation principle);
- $p_T(\mu) > 35 \text{ GeV}$ and $p_T(\tau_h) > 100 \text{ GeV}$ (PPS acceptance);

5.3 $\tau_h \tau_h$ final state

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- $|\eta| < 2.3$ for both the muon and the τ_h ;
- at least 1 reconstructed proton per PPS arm.

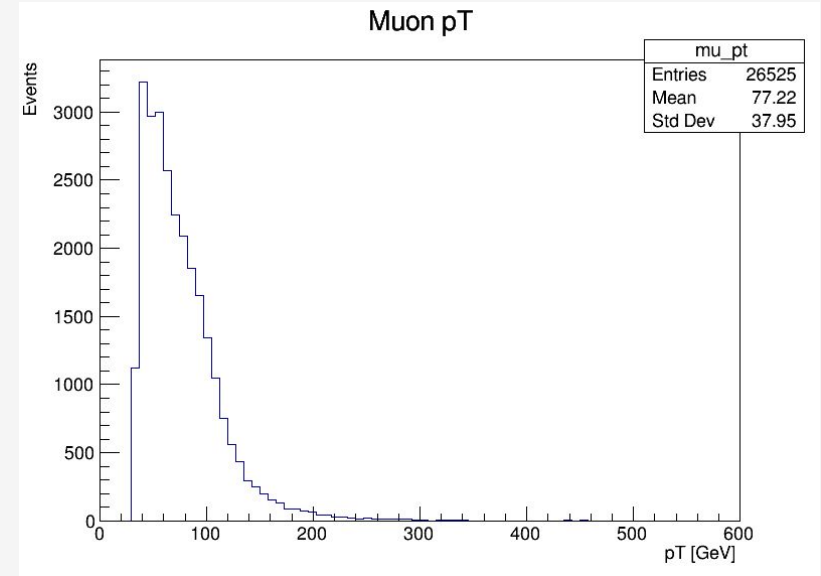
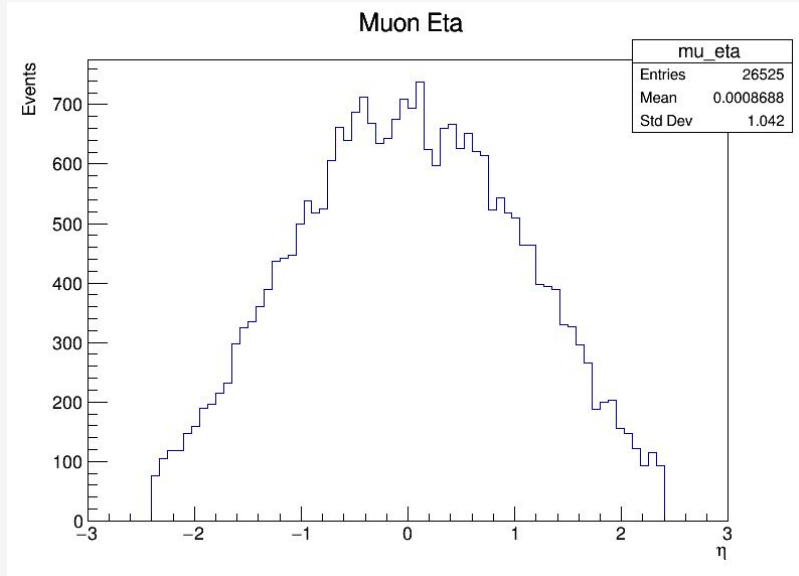
Please note that in this preliminary event selection we do not mention any cut based on the relations $M_X^2 = s \cdot \xi_1 \xi_2$ and $Y_X = 0.5 \cdot \log(\xi_1/\xi_2)$. These conditions will be taken into account later (MVA tool).

The number of events (normalised to 54.9 fb^{-1}) surviving the cuts at different selection steps are summarized in Table 8. Final numbers take into account protons corrections. Also in this case, the QCD background is estimated using a data-driven approach (method described in Sec. 3).

cut		
Trigger	<i>HLT_IsoMu24</i>	phase0
one tau and one muon		phase0
muon cuts	<i>pt > 35, η < 2.4</i>	phase1
tau cuts	<i>pt > 100, η < 2.4</i>	phase1
opposite sign		phase1
SFs		phase1

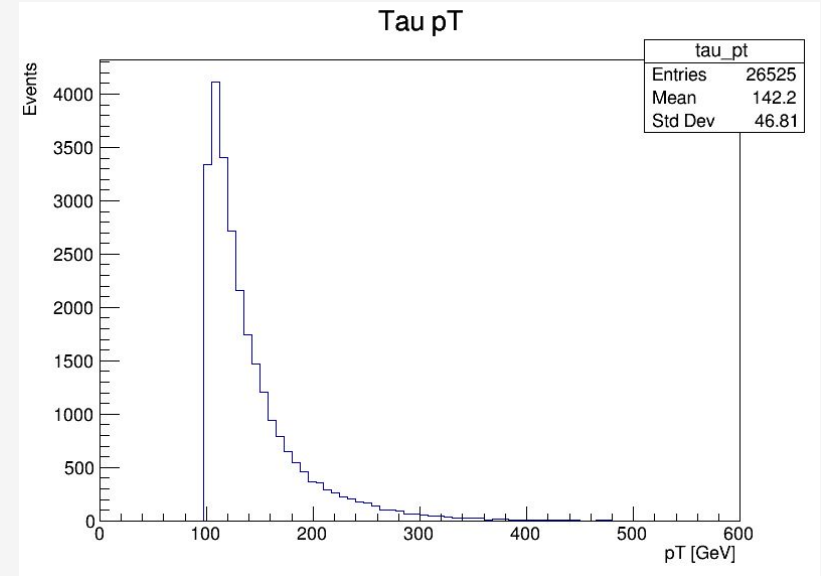
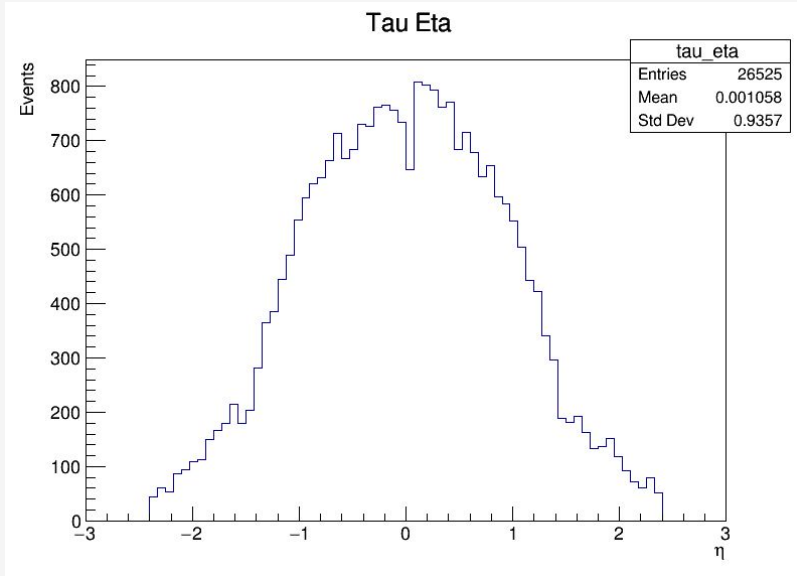
MuTau channel -> ttjets background; no pileup protons

- one isolated tau and one isolated muon (opposite charge)
- tau $p_T > 100 \text{ GeV}$; mu $p_T > 35 \text{ GeV}$
- at least one proton per pps arm



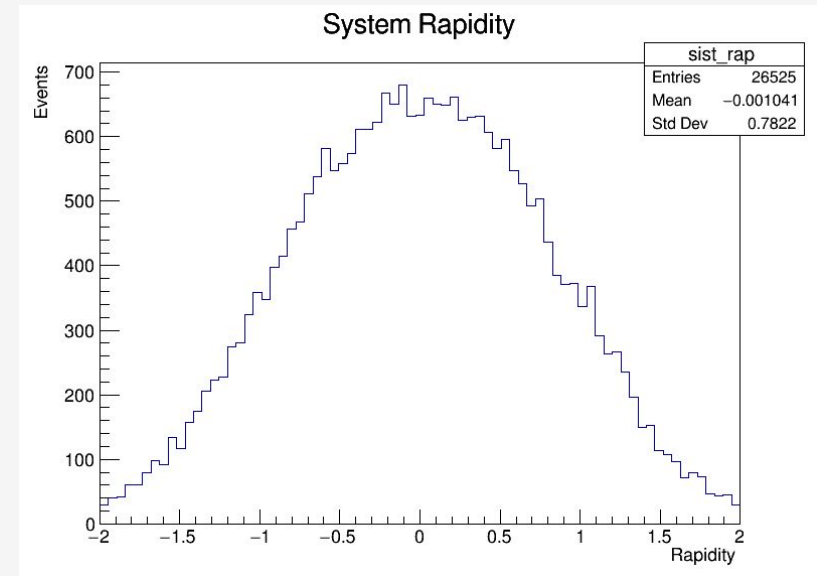
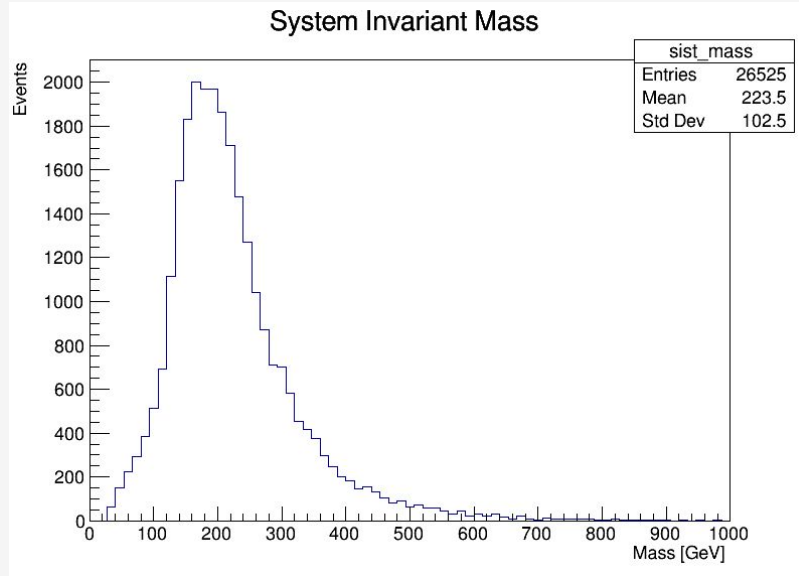
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MuTau channel -> ttjets background; no pileup protons

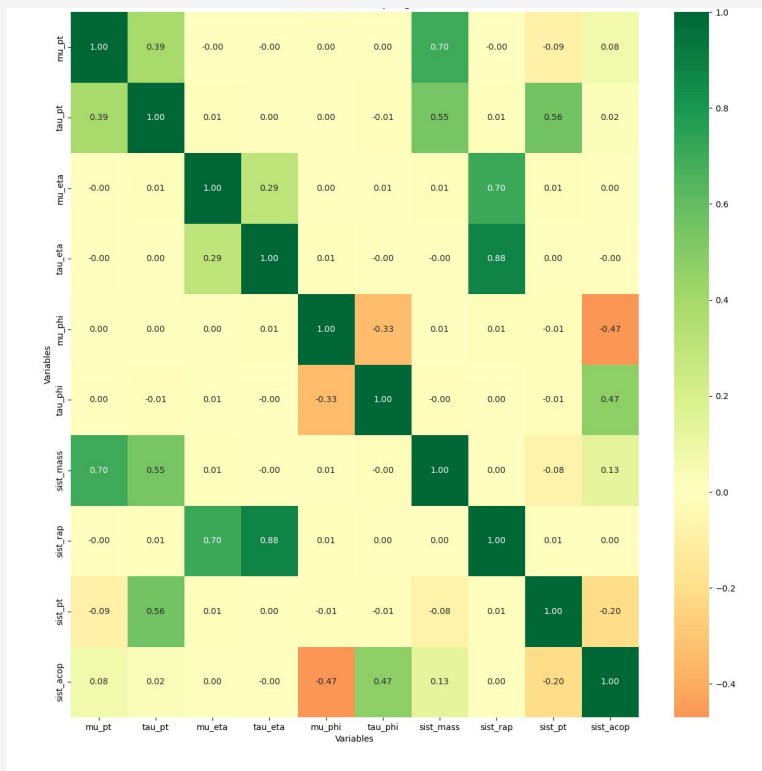
- one isolated tau and one isolated muon (opposite charge)
- tau $p_t > 100 \text{ GeV}$; mu $p_t > 35 \text{ GeV}$
- at least one proton per pps arm



03

Correlation between variables

ttjets phase1 -> pileup protons not yet introduced



Next:

- Run phase1 for DY, QCD
- merge and introduce pileup protons
- Process signal sample
- plot and analyse bdt discriminators
- plot control regions

Input distributions to the BDT discriminator

IN PROGRESS

- Invariant mass of the central system
→ The total mass of all particles in the central detector, calculated from their 4-momenta.
- Rapidity of the central system
→ Describes how forward or backward the central system is moving along the beam axis.
- Mass difference
→ Difference between the central system mass and the mass reconstructed using the forward protons (CMS - PPS).
- MET (Missing Transverse Energy)
→ Represents undetected particles.
- Transverse momentum of the central system
→ Momentum of the central system perpendicular to the beam axis.
- Rapidity matching
→ Comparison of rapidity measured centrally..
- Transverse momentum of hadronic tau
→ Momentum perpendicular to the beam of the hadronically decaying tau lepton.
- Acoplanarity of central system
→ Measures how "non-back-to-back" the two leading tau candidates are; zero means perfect balance.